

ASSESSMENT TEST: SOFTWARE TESTER (QA)

Task Management Application – Test Case Design & Bug Identification

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Application Understanding

The application is a Task Management Application with User Registration/Login. After login, users can create, view, edit and delete tasks. Task data is stored in a database and displayed in a list. The application is assumed to be close to production release and has no existing test documentation.

1. Test Case Design – Registration

TC ID	Test Scenario / Steps	Expected Result	Type
REG-01	Enter valid name, email and password and submit registration.	Account should be created successfully.	Positive
REG-02	Register using an email that already exists.	Registration should fail with a clear duplicate-email message.	Negative
REG-03	Enter an invalid email format.	Validation message should be displayed.	Negative
REG-04	Leave one or more required fields empty and submit.	Required-field validation should be shown.	Negative
REG-05	Enter a password shorter than the allowed minimum.	Registration should be rejected with a password validation message.	Edge
REG-06	Enter input at the maximum allowed length.	Valid input should be accepted without UI/database issues.	Edge

2. Test Case Design – Login

TC ID	Test Scenario / Steps	Expected Result	Type
LOG-01	Login with valid registered email and password.	User should be logged in and taken to the task area.	Positive
LOG-02	Enter correct email with an incorrect password.	Login should fail with an appropriate error message.	Negative
LOG-03	Login with an unregistered email.	Login should fail and no task data should be shown.	Negative
LOG-04	Leave email/password blank and submit.	Required-field validation should be displayed.	Negative
LOG-05	Enter an invalid email format.	Email validation should be displayed.	Negative
LOG-06	Enter unusually long or unexpected input in login fields.	Application should handle input safely without crashing.	Edge

3. Test Case Design – Task CRUD Operations

TC ID	Test Scenario / Steps	Expected Result	Type
TASK-01	Create a task using valid details.	Task should be saved and displayed in the task list.	Positive
TASK-02	Create a task with a required field left blank.	Task should not be created; validation should be shown.	Negative
TASK-03	Create several tasks one after another.	All valid tasks should appear correctly in the list.	Positive
TASK-04	Edit an existing task and save valid changes.	Updated task should be displayed and saved in the database.	Positive
TASK-05	Edit a task with invalid/blank data.	Invalid changes should not be saved.	Negative
TASK-06	Delete an existing task.	Task should disappear from the list and database.	Positive
TASK-07	Start deleting a task and cancel the action, if confirmation exists.	Task should remain unchanged.	Edge
TASK-08	Create/edit/delete a task and refresh the page.	The list should match the latest database state.	Edge

4. Input Validation & Error Handling

TC ID	Test Scenario / Steps	Expected Result	Type
VAL-01	Submit forms with all required fields blank.	Clear validation messages should be displayed.	Negative
VAL-02	Enter only spaces in name/task fields.	Input should be rejected or trimmed correctly.	Edge
VAL-03	Enter special characters in task fields.	Application should handle the input safely and correctly.	Edge
VAL-04	Enter extremely long task text.	Application should not crash or break the layout.	Edge
VAL-05	Simulate/encounter a database or server failure while saving.	User should receive a meaningful error message and data should not be silently lost.	Negative
VAL-06	Access task pages after logout/session expiry.	User should be redirected to login and protected data should not be accessible.	Negative

5. Bug Identification

The following are potential bugs/risk areas identified without executing the application, based on the given requirements and production-release context.

Bug ID	Potential Bug / Risk Area	Severity	Reason / Impact
BUG-01	Duplicate email may be accepted during registration.	Major	Can create duplicate accounts and inconsistent user records.
BUG-02	Weak or invalid passwords may be accepted.	Major	Can reduce account security and increase unauthorized-access risk.
BUG-03	A logged-out/unauthorized user may access task data.	Critical	Could expose private user information.
BUG-04	Tasks may be created with missing or invalid required data.	Major	Produces incomplete or unusable task records.
BUG-05	Deleted task may remain in the database.	Major	Task list and database can become inconsistent.
BUG-06	Edited task may update only the UI but not the database.	Major	Old data may reappear after refresh.
BUG-07	Database/server failure may not be handled properly.	Major	Users may lose data or receive no useful feedback.
BUG-08	Very long or unexpected input may break the UI.	Minor	Can cause poor user experience or layout issues.
BUG-09	Session expiry/logout may not properly restrict access.	Critical	Unauthorized users may continue viewing protected data.
BUG-10	Tasks may be incorrectly mapped between users.	Critical	One user's tasks could become visible to another user, causing a serious privacy issue.

6. Testing Priorities Before Production Release

- Authentication and authorization should be tested first, especially protection of user-specific task data.
- All CRUD operations should be verified for both UI behavior and database consistency.
- Registration/login validation and error messages should be checked for invalid and edge inputs.
- Session expiry and logout behavior should be tested to prevent unauthorized access.
- Database/server failure scenarios should be handled gracefully with clear user feedback.

Conclusion

The highest-risk areas are authentication, authorization, user-task data isolation, CRUD/database consistency, input validation and error handling. Testing should cover positive, negative and edge cases before the application is released to production.